

2025-11-07 - Valentini (12)

Previous lecture: [2025-11-06 - Valentini \(11\)](#)

Next lecture: [2025-11-14 - Servidio \(13\)](#)

Energy paradox

Vlasov system preserves both energy and entropy.

Landau damping drains energy from the electric field.

Where does the energy go?

How can we get it back reversing time since entropy is conserved?

It cannot go into thermal energy because we don't have collisions.

What we did is the following

$$\tilde{f}_{\alpha k}(\bar{v}, p) = \frac{f_{\alpha k}(\bar{v}, t=0)}{p + i(\bar{k} \cdot \bar{v})} + \frac{i \frac{q_\alpha}{m_\alpha} \tilde{\phi}_k(p) \bar{k} \cdot \nabla_{\bar{v}} f_{\alpha 0}}{p + i(\bar{k} \cdot \bar{v})}$$

$$k^2 \tilde{\phi}_k(p) = 4\pi \sum_{\alpha} q_{\alpha} \int_{-\infty}^{\infty} \tilde{f}_{\alpha k}(\bar{v}, p) d^3 v$$

and (using inverse Laplace Transform)

$$f_{\alpha k}(\bar{v}, t) = \int_{p_0 - i\infty}^{p_0 + i\infty} \tilde{f}_{\alpha k}(\bar{v}, p) e^{pt} \frac{dp}{2\pi i}$$

There is for this integral a group of poles which are also poles of $\tilde{\phi}_k(p)$

Using the same rotation as before

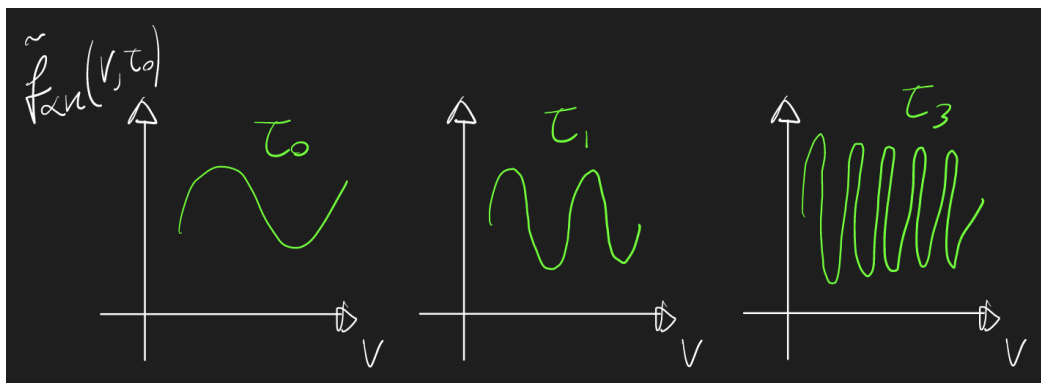
$$f_{\alpha k}(\bar{v}, t) = \sum_j f_{\alpha k}(\bar{v}) \exp(-i\omega_j t) + f_{\alpha \beta}(\bar{v}) \exp(-i\bar{k} \cdot \bar{v} t)$$

Where in the sum we take care of the poles of $\tilde{\phi}$.

This first term is the Landau damping contribution of $\tilde{\phi}_k(t)$.

However $f_{\alpha k}(\bar{v}, t=0)$ has an extra pole, the second term, which is the ballistic contribution. This term oscillates and creates more and more wavelengths in space as time advances.

PIC it oscillates more and more



The ballistic term stores the energy lost by Landau damping.

If we plug this $f_{\alpha k}(\bar{v}, t)$ in the previous expression for $k^2 \tilde{\phi}_k(p)$, why don't we see the ballistic pole in this solution?

(before we used a different method to reach a solution for this quantity)

$$\begin{aligned}
k^2 \phi_n(t) &= 4\pi \sum_{\alpha} q_{\alpha} \int_{-\infty}^{\infty} f_{\alpha \bar{k}}(\bar{v}, t) d^3 v = \\
&= 4\pi \sum_{\alpha} q_{\alpha} \left[\int_{-\infty}^{\infty} \sum_j f_{\alpha k}(\bar{v}) \exp(-i\omega_j t) d^3 v + \int_{-\infty}^{\infty} f_{\alpha \beta}(\bar{v}) \exp(-i\bar{k} \cdot \bar{v} t) d^3 v \right]
\end{aligned}$$

We try to solve the second integral by applying a rotation such that $\bar{k} \parallel \hat{u}$

$$\begin{aligned}
\mathcal{I} &\equiv \int_{-\infty}^{\infty} f_{\alpha \beta}(u) \exp(-ikut) du = \\
&= \int_{-\infty}^{\infty} \frac{n_0}{\sqrt{2\pi} v_{th e}} \exp\left(-\frac{u^2}{2v_{th e}^2}\right) \exp(-ikut) du = \\
&= \frac{n_0}{\sqrt{2\pi} v_{th e}} \int_{-\infty}^{\infty} \exp\left[-\frac{1}{2}\left(\frac{u}{v_{th e}} + ikv_{th e} t\right)^2\right] \exp\left(-\frac{k^2 v_{th e}^2 t^2}{2}\right) du
\end{aligned}$$

Where in the last passage we used square completion method.

We substitute

$$y = \frac{u}{v_{th e}} + ikv_{th e} t$$

and

$$\begin{aligned}
\mathcal{I} &= \frac{n_0 \cancel{v_{th e}}}{\sqrt{2\pi} \cancel{v_{th e}}} \exp\left(-\frac{k^2 v_{th e}^2 t^2}{2}\right) \int_{-\infty}^{\infty} \exp\left(-\frac{y^2}{2}\right) dy = \\
&= \frac{n_0}{\cancel{\sqrt{2\pi}}} \exp\left(-\frac{k^2 v_{th e}^2 t^2}{2}\right) \cdot \cancel{\sqrt{2\pi}} = \\
&= n_0 \exp\left(-\frac{k^2 v_{th e}^2 t^2}{2}\right)
\end{aligned}$$

We see **phase mixing**

$\mathcal{I} \sim \exp(-t^2)$ so we lose its contribution when we neglect the transient as we did before.

HERE so the energy paradox is that the energy can only be recovered with a second order term which we neglected?

Plasma wave echo theory

O'Neil, Ceuld, Malbourg, 1965

Here we show a toy model of their theory.

Perturbation of the electric field, one-dimensional system

$$\begin{aligned}
E_1(x, t) &= E_1 \exp(-ik_1 x - i\omega_1 t) = \\
&= E_1 \exp(-ik_1 x - i\omega_R^{(1)} t + i\omega_I^{(1)} t) \\
f_1(x, v, t) &= f_1(v) \exp(-ik_1 x - i\omega_R^{(1)} t + i\omega_I^{(1)} t) + \\
&\quad + f_{1\beta}(v) \exp(-ik_1 x + ik_1 vt)
\end{aligned}$$

Where the second term is the ballistic term and is not damped.

We say that the delay time between the first and the second perturbation in the electric field is

$$\tau > \omega_I^{-1}$$

the second perturbation is

$$E_2(x, t) = E_2 \exp \left(-ik_2 x - i\omega_R^{(2)} t + i\omega_I^{(2)} t \right)$$

$$f_2(x, v, t) = f_2(v) \exp \left[-ik_2 x - i\omega_R^{(2)} (t - \tau) + i\omega_I^{(2)} (t - \tau) \right] +$$

$$+ f_{2\beta}(v) \exp \left[-ik_2 x + ik_2 v (t - \tau) \right]$$

The two ballistic terms, which survive for a long time, couple together.

They can make constructive interference, which would be a second-order term.

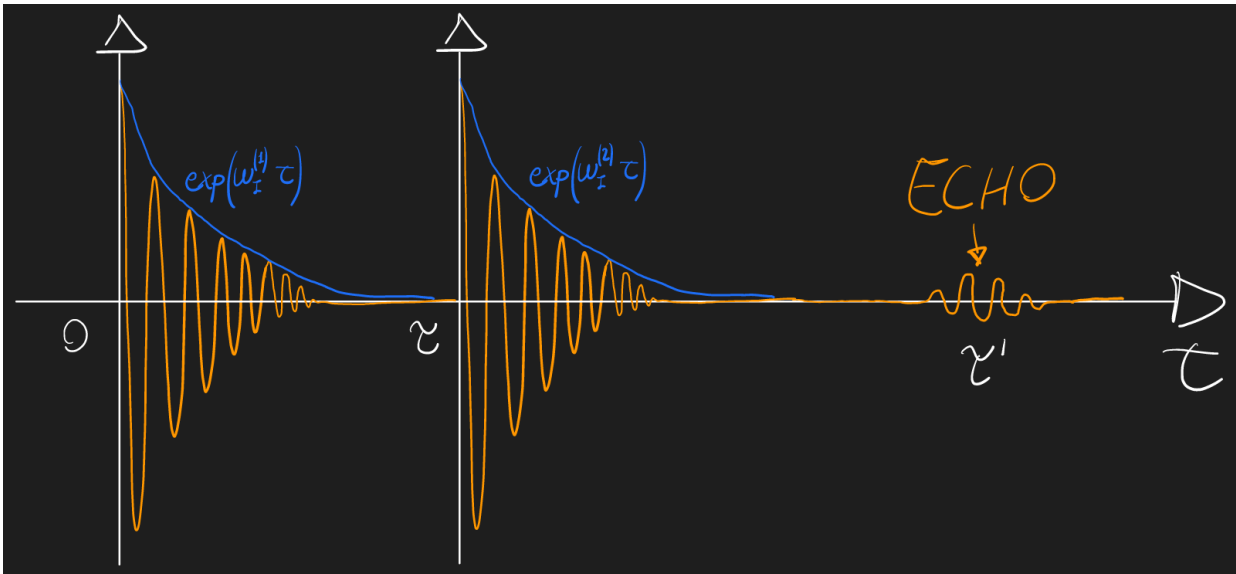
$$\mathcal{I} = f_{1\beta}(v) f_{2\beta}(v) \exp[i(k_1 - k_2)x - i(k_2 - k_1)vt + ik_2 v \tau]$$

The contribution of this nonlinear term in the potential

$$\tau' = \frac{k_2}{k_2 - k_1} \tau$$

at τ' the exponent in the previous expression is not dependent on velocity and its contribution is visible.

PIC battimenti - Echo



We also have 2nd and 3rd echos, but they are much more difficult to observe.

Energetics in wave-particles interaction

$$\frac{\partial}{\partial t} f(x, v, t) + v \frac{\partial}{\partial x} f(x, v, t) - \frac{eE}{m_e} \frac{\partial}{\partial v} f(x, v, t) = 0$$

$$\frac{\partial E(x, t)}{\partial x} = 4\pi e \left[n_0 - \int_{-\infty}^{\infty} f(x, v, t) dv \right]$$

with

$$f(x, v, t) = f_0(v) + f_1(x, v, t)$$

$$f_0(v) = \frac{n_0 m_e^{1/2}}{(2\pi k_B T_{e0})^{1/2}} \exp\left(-\frac{m_e}{2k_B T_{e0}} v^2\right)$$

$$E(x, t) = E_1(x, t)$$

$$n_{i0} = n_{e0} = n_0$$

And the linearized Vlasov equation is

$$\frac{\partial}{\partial t} f_1(x, v, t) + v \frac{\partial}{\partial x} f_1(x, v, t) - \frac{eE}{m_e} \frac{\partial}{\partial v} f_0(v) = 0$$

We also want to linearize the Poisson equation, so we write

$$E_1(x, t) = E_0 \exp(\omega_I t) \sin(kx - \omega_R t)$$

using the approximation of $|\omega_I/\omega_R| \ll 1$ we get

$$E_1(x, t) \approx E_0 \sin(kx - \omega_R t)$$

The poisson equation at time $t = 0$ will be

$$\begin{aligned} \frac{\partial E_1(x, 0)}{\partial x} &= 4\pi e \left[\cancel{n_0} - \cancel{\int_{-\infty}^{\infty} f_0(v) dv} - \int_{-\infty}^{\infty} f_1(x, v, 0) dv \right] = \\ &= -4\pi e \int_{-\infty}^{\infty} f_1(x, v, 0) dv \end{aligned}$$

(the first two terms in the parenthesis are both n_0).

Now we assume

$$f_1(x, v, 0) = f_0(v) g(x)$$

and

$$\begin{aligned} \frac{\partial E_1(x, 0)}{\partial x} &= -4\pi e \int_{-\infty}^{\infty} f_0(v) g(x) dv = \\ &= -4\pi e g(x) n_0 \end{aligned}$$

At $t = 0$ we also have

$$E_1(x, t) = E_0 \sin(kx)$$

so using this into the previous expression yields

$$kE_0 \cos(kx) = -4\pi e g(x) n_0 \quad \implies \quad g(x) = -\frac{kE_0}{4\pi e n_0} \cos(kx)$$

we write it as

$$g(x) = \beta_k \cos(kx), \quad \text{with } \beta_k = -\frac{kE_0}{4\pi e n_0}$$

So

$$\begin{cases} E_1(x, 0) = E_0 \sin(kx) \\ f_1(x, v, 0) = f_0(v) \beta_k \cos(kx) \end{cases}$$

We use the **method of characteristics**

$$\frac{\partial f}{\partial t} + c \frac{\partial f}{\partial x} = 0$$

with $f = f(x, t)$ and the following Initial Condition

$$f(x, 0) = f_0(x)$$

We write

$$\frac{d}{dt} f = \frac{\partial f}{\partial t} + \frac{\partial x}{\partial t} \frac{\partial f}{\partial x}$$

which, for the case $\partial x / \partial t = c$ becomes

$$\frac{\partial f}{\partial t} + c \frac{\partial f}{\partial x} = 0$$

so

$$\begin{aligned} \left. \frac{df}{dt} \right|_{\text{orbit}} &= 0 \\ \int_0^t \frac{df}{dt'} dt' = 0 &\implies \frac{dx}{dt} = c \implies x(t) = x_0 + ct \end{aligned}$$

and also

$$\begin{aligned} \int_0^t \frac{d}{dt'} (f(x(t'), t')) dt' &= 0 \\ \implies f(x(t'), t') \Big|_0^t &= 0 \\ \implies f(x(t), t) - f(x(0), 0) &= 0 \\ \implies f(x(t), t) = f(x_0, t_0) &\stackrel{(IC)}{=} f_0(x_0) = f_0(x - vt) \end{aligned}$$

In our case

$$\frac{\partial f_1}{\partial t} + v \frac{\partial f_1}{\partial x} = \frac{eE}{m_e} \frac{\partial f_0}{\partial v}$$

with $f_1 = f_1(x, v, t)$

$$\frac{df_1}{dt} = \frac{\partial f_1}{\partial t} + \frac{\partial x}{\partial t} \frac{\partial f_1}{\partial x} + \frac{\partial v}{\partial t} \frac{\partial f_1}{\partial v}$$

with

$$\frac{\partial x}{\partial t} = v, \quad \frac{\partial v}{\partial t} = -\frac{eE_1}{m_e}$$

and

$$0 = \left. \frac{df}{dt} \right|_{\text{orbit}} = \frac{\partial f_1}{\partial t} + v \frac{\partial f_1}{\partial x} - \frac{eE_1}{m_e} \frac{\partial f_1}{\partial v} \approx \frac{\partial f_1}{\partial t} + v \frac{\partial f_1}{\partial x} - \frac{eE_1}{m_e} \frac{\partial f_0}{\partial v}$$

where we linearized and neglected the non linear term.

Finally we have

$$\frac{df_1}{dt} = \frac{eE_1}{m_e} \frac{\partial f_0}{\partial v}$$

so

$$x(t) = x_0 + vt, \quad v = \text{const}$$

$$\int_0^t \frac{df_1(x, v, t')}{dt'} dt' = \int_0^t \frac{eE_1}{m_e} \frac{\partial f_0(v)}{\partial v} dt'$$

The first term is

$$\begin{aligned} \int_0^t \frac{df_1(x, v, t')}{dt'} dt' &= f_1(x, v, t) - f_1(x, v, 0) = \\ &= f_1(x, v, t) - f_1(x_0, 0) = \\ &= f_1(x, v, t) - f_1(x - vt, 0) = \\ &= f_1(x, v, t) - f_0 \beta_k \cos[k(x - vt)] \end{aligned}$$

and substituting it we get

$$\begin{aligned} f_1(x, v, t) &= f_0 \beta_k \cos[k(x - vt)] + \int_0^t \frac{eE_1}{m_e} \frac{\partial f_0(v)}{\partial v} dt' = \\ &= f_0 \beta_k \cos[k(x - vt)] + \frac{e}{m_e} \frac{\partial f_0}{\partial v} \int_0^t E_1(x_0 + vt', t') dt' = \\ &= f_0 \beta_k \cos[k(x - vt)] + \frac{e}{m_e} \frac{\partial f_0}{\partial v} \int_0^t E_0 \sin(kx_0 + kv t' - \omega_R t') dt' = \\ &= f_0 \beta_k \cos[k(x - vt)] + \underbrace{\frac{e}{m_e} \frac{\partial f_0}{\partial v} \frac{E_0}{kv - \omega_R} [-\cos(kx_0 + kv t' - \omega_R t')]_0^t}_{*} \end{aligned}$$

and

$$\begin{aligned} * &= \frac{e}{m_e} \frac{\partial f_0}{\partial v} \frac{E_0}{kv - \omega_R} [-\cos(kx_0 + kv t' - \omega_R t')]_0^t = \\ &= \frac{e}{m_e} \frac{\partial f_0}{\partial v} \frac{E_0}{kv - \omega_R} [\cos(kx_0) - \cos(kx_0 + kv t - \omega_R t)] = \\ &= \frac{e}{m_e} \frac{\partial f_0}{\partial v} \frac{E_0}{kv - \omega_R} [\cos(kx_0) - \cos(kx - \cancel{kv t} + \cancel{kv t} - \omega_R t)] = \\ &= \frac{e}{m_e} \frac{\partial f_0}{\partial v} \frac{E_0}{kv - \omega_R} [\cos[k(x - vt)] - \cos(kx - \omega_R t)] = \\ &= \frac{e}{m_e} \frac{\partial f_0}{\partial v} \frac{E_0}{k} \cdot \frac{\cos[k(x - vt)] - \cos(kx - \omega_R t)}{v - \omega_R/k} \end{aligned}$$

and finally

$$f_1(x, v, t) = f_0 \beta_k \cos[k(x - vt)] + \frac{e}{m_e} \frac{\partial f_0}{\partial v} \frac{E_0}{k} \cdot \frac{\cos[k(x - vt)] - \cos(kx - \omega_I t)}{v - \omega_R/k}$$

(we will continue from here)

In linear vlasov we neglect nonlinear term

$$\frac{eE}{m_e} \frac{\partial f_1}{\partial v}$$

so lin. approx is broken when

$$\left| \frac{\partial f_1}{\partial v} \right| \sim \left| \frac{\partial f_0}{\partial v} \right|$$

which happens at time τ

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